



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

Student Academic Performance Analysis and Grade Distribution

Project Domain

[Excel]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business
Intelligence

by:



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3. Problem Statement

Educational institutions generate large amounts of student academic data every semester. However, this data is often underutilized when it comes to understanding student performance patterns and identifying areas for improvement. Without proper analysis, it becomes difficult for educators to evaluate student progress, detect weak subject areas, and improve teaching strategies.

The Student Academic Performance Analysis and Grade Distribution project aims to analyze student marks across multiple subjects, calculate percentages, assign grades, and visualize the distribution of academic performance. By applying data analysis techniques and visualization tools, the project provides meaningful insights that can help educators and administrators make informed decisions to enhance student learning outcomes.

4. Objectives

The main objectives of this project are:

- To collect and organize student academic data including subject-wise marks.
- To calculate total marks, percentage, grade, and result for each student.
- To analyze the academic performance of students across different subjects.
- To identify grade distribution among students.
- To visualize performance trends using charts and graphs.
- To provide insights that help educators understand strengths and weaknesses in student performance.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	Data Collection & Exploration	The first step of the project is collecting the dataset containing academic records of students. The dataset includes information such as: Student ID, Student Name, Gender, Department, Group, Marks in subjects (English, Mathematics, Economics, Physics, Biology, Chemistry, Science)

		The dataset contains 500 student records. Initial exploration is performed to understand the structure of the data and identify patterns or missing values.
2	Data Cleaning & Preprocessing	Before performing analysis, the dataset must be cleaned and prepared. Tasks involved include: Removing duplicate records, Handling missing values if any, Standardizing column names and formats, Verifying marks are within valid ranges (0–100) Calculating derived fields such as: Obtained Marks, Total Marks, Percentage, Grade, Result, Data preprocessing ensures that the dataset is accurate and ready for analysis
3	Analysis & AI Integration	After preprocessing, analytical methods are applied to extract meaningful insights. Key analysis includes: ● Calculating average marks per subject ● Identifying top-performing students ● Determining pass and fail rates ● Evaluating grade distribution ● Comparing performance across departments or groups ● AI-based tools or data analytics techniques may be used to: ● Predict student performance trends ● Identify subjects where students perform poorly ● Provide recommendations for academic improvement
4	Visualization / Output	Data visualization helps present complex data in a simple and understandable format. Various visualization techniques include: ● Bar charts – Subject-wise average marks ● Pie charts – Grade distribution ● Histograms – Percentage distribution ● Line charts – Performance trends Tools that can be used include:

		● Excel ● Python (Pandas, Matplotlib, Seaborn) ● Power BI / Tableau ● These visual outputs help educators quickly interpret the data.
5	Documentation & Submission	The final stage of the project involves documenting the entire work. The documentation includes: ● Introduction and problem statement ● Objectives of the study ● Methodology used for analysis ● Data analysis results and visualizations ● Conclusion and recommendations ● The final report and dataset are then submitted as part of the project deliverables.

6. Resources and Timeline

A. Hardware & Software Requirements

Hardware

- Computer / Laptop
- Minimum 4GB RAM
- Internet connectivity

Software

- Microsoft Excel
- Python (Jupyter Notebook / Google Colab)
- Power BI or Tableau (for visualization)
- Microsoft Word or Google Docs for documentation

B. Dataset

The dataset used in this project consists of 500 student academic records containing the following attributes:

1. Student ID

2. Student Name
3. Gender
4. Department
5. Group
6. English Marks
7. Mathematics Marks
8. Economics Marks
9. Physics Marks
10. Biology Marks
11. Chemistry Marks
12. Science Marks
13. Obtained Marks
14. Total Marks
15. Percentage
16. Grade
17. Result

The dataset may be generated manually, collected from institutional records, or simulated for academic analysis.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	1 Day
2	Data cleaning, preprocessing, and validation / Prompt design	1 Day
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	2 Days
4	Dashboard / output build, testing, and review	1 Day
5	Report writing, GitHub upload, and video recording and submission	2 Days

7. Expected Outcomes / Conclusion

The project will produce a detailed analysis of student academic performance and provide insights into grade distribution. It will help identify top-performing students, subject difficulty levels, and overall pass rates.

Through data visualization and analysis, educators can better understand student performance trends and make data-driven decisions to improve teaching strategies.

Overall, this project demonstrates how data analytics and visualization techniques can be applied in the education sector to enhance academic evaluation and student success.

[Add any outcome specific to your project domain here — for example: a prompt library, a published Power BI dashboard, a marketing content portfolio, or health metrics visualizations.]

8. References

- Microsoft Excel Documentation
- Python Pandas Library Documentation
- Matplotlib & Seaborn Visualization Libraries
- Educational Data Mining Research Papers
- Online tutorials from platforms such as Coursera, Kaggle, and DataCamp
- *[Add your project-specific references here: research papers, GitHub links, YouTube tutorials, blogs, or any other material consulted.]*